

Simulation of Assem4

Date: Thursday, April 9, 2026
Designer: Solidworks
Study name: assembly_1_static
Analysis type: Static

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Description
No Data



Assumptions



Model Information





Model name: Assem4
Current Configuration: Default

Solid Bodies

Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
 B6702ZZ.sldprt	Solid Body	Mass:0.00515218 kg Volume:6.44023e-07 m ³ Density:8,000 kg/m ³ Weight:0.0504914 N	C:\Users\ENG-HASSAN\OneDrive\Documents\Solidworks\assembly\B6702ZZ.sldprt Mar 29 01:12:41 2026
 B6804ZZ.sldprt	Solid Body	Mass:0.0270741 kg Volume:3.38426e-06 m ³ Density:8,000 kg/m ³ Weight:0.265326 N	C:\Users\ENG-HASSAN\OneDrive\Documents\Solidworks\assembly\B6804ZZ.sldprt Mar 29 01:12:41 2026
 B6809ZZ.sldprt	Solid Body	Mass:0.0557469 kg Volume:6.96836e-06 m ³ Density:8,000 kg/m ³ Weight:0.546319 N	C:\Users\ENG-HASSAN\OneDrive\Documents\Solidworks\assembly\B6809ZZ.sldprt Mar 29 01:12:41 2026
 B6809ZZ.sldprt	Solid Body	Mass:0.0557469 kg Volume:6.96836e-06 m ³ Density:8,000 kg/m ³ Weight:0.546319 N	C:\Users\ENG-HASSAN\OneDrive\Documents\Solidworks\assembly\B6809ZZ.sldprt Mar 29 01:12:41 2026
 B6902ZZ.sldprt	Solid Body	Mass:0.0224059 kg Volume:2.80074e-06 m ³ Density:8,000 kg/m ³ Weight:0.219578 N	C:\Users\ENG-HASSAN\OneDrive\Documents\Solidworks\assembly\B6902ZZ.sldprt Mar 29 01:12:41 2026
MUEN15.sldprt	Solid Body	Mass:0.011334 kg Volume:1.41675e-06 m ³ Density:8,000 kg/m ³	C:\Users\ENG-HASSAN\OneDrive\Docum



		Weight:0.111073 N	ents\Solidworks\assembly\ MUEN15.sldprt Mar 29 01:12:45 2026
MUEN45.sldprt 	Solid Body	Mass:0.0670728 kg Volume:8.38411e-06 m ³ Density:8,000 kg/m ³ Weight:0.657314 N	C:\Users\ENG- HASSAN\OneDrive\Docum ents\Solidworks\assembly\ MUEN45.sldprt Mar 29 01:12:45 2026
Split Line1 	Solid Body	Mass:0.137864 kg Volume:1.7233e-05 m ³ Density:8,000 kg/m ³ Weight:1.35107 N	C:\Users\ENG- HASSAN\OneDrive\Docum ents\Solidworks\assembly\ blenderaugerlong.SLDPRT Mar 29 01:12:42 2026
M6 Tapped Hole1 	Solid Body	Mass:0.0931969 kg Volume:1.16496e-05 m ³ Density:8,000 kg/m ³ Weight:0.91333 N	C:\Users\ENG- HASSAN\OneDrive\Docum ents\Solidworks\assembly\ blenderaugershortonly.SL DPRT Mar 29 01:12:42 2026
M6 Tapped Hole1 	Solid Body	Mass:0.86854 kg Volume:0.000108567 m ³ Density:8,000 kg/m ³ Weight:8.51169 N	C:\Users\ENG- HASSAN\OneDrive\Docume nts\Solidworks\assembly\bl enderseat.SLDPRT Mar 29 01:12:42 2026
Chamfer2 	Solid Body	Mass:0.293127 kg Volume:3.66409e-05 m ³ Density:8,000 kg/m ³ Weight:2.87265 N	C:\Users\ENG- HASSAN\OneDrive\Docume nts\Solidworks\assembly\bl ocker.SLDPRT Mar 29 01:12:42 2026
Extrude2 	Solid Body	Mass:0.0720602 kg Volume:9.00752e-06 m ³ Density:8,000 kg/m ³ Weight:0.70619 N	C:\Users\ENG- HASSAN\OneDrive\Docume nts\Solidworks\assembly\bl ocker2.SLDPRT Mar 29 01:12:42 2026
M4 Tapped Hole1 	Solid Body	Mass:1.21131 kg Volume:0.000151413 m ³ Density:8,000 kg/m ³ Weight:11.8708 N	C:\Users\ENG- HASSAN\OneDrive\Docume nts\Solidworks\assembly\ou tbearingseat.SLDPRT Mar 29 01:12:45 2026
Extrude6 	Solid Body	Mass:0.320953 kg Volume:4.01191e-05 m ³ Density:8,000 kg/m ³ Weight:3.14534 N	C:\Users\ENG- HASSAN\OneDrive\Docume nts\Solidworks\assembly\sh aft.SLDPRT Mar 29 01:12:45 2026
Extrude3	Solid Body	Mass:0.836782 kg Volume:0.000104598 m ³	C:\Users\ENG- HASSAN\OneDrive\Docum



		Density:8,000 kg/m ³ Weight:8.20046 N	ents\Solidworks\assembly\ sleeve7.SLDPRT Mar 29 01:12:45 2026
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Study Properties

Study name	assembly_1_static
Analysis type	Static
Mesh type	Solid Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Large displacement	On
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off
Result folder	SOLIDWORKS document (G:\assembly\assembly)

Units

Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m ²

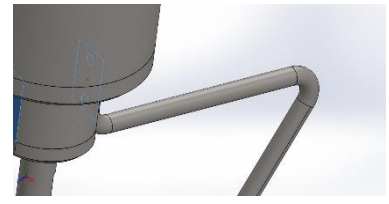


Material Properties

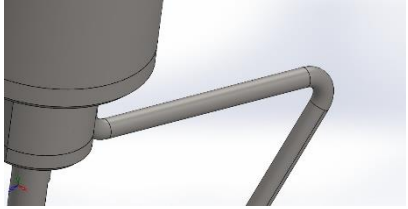
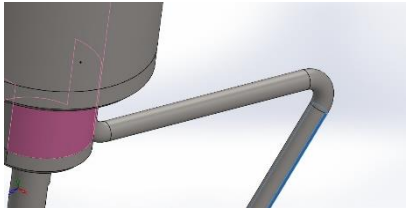
Model Reference	Properties	Components
	Name: AISI 304 Model type: Linear Elastic Isotropic Default failure criterion: Max von Mises Stress Yield strength: 2.06807e+08 N/m ² Tensile strength: 5.17017e+08 N/m ² Elastic modulus: 1.9e+11 N/m ² Poisson's ratio: 0.29 Mass density: 8,000 kg/m ³ Shear modulus: 7.5e+10 N/m ² Thermal expansion coefficient: 1.8e-05 /Kelvin	SolidBody 1(B6702ZZ.sldprt)(B6702ZZ-1), SolidBody 1(B6804ZZ.sldprt)(B6804ZZ-1), SolidBody 1(B6809ZZ.sldprt)(B6809ZZ-1), SolidBody 1(B6809ZZ.sldprt)(B6809ZZ-2), SolidBody 1(B6902ZZ.sldprt)(B6902ZZ-1), SolidBody 1(MUEN15.sldprt)(MUEN15-1), SolidBody 1(MUEN45.sldprt)(MUEN45-1), SolidBody 1(Split Line1)(blenderaugerlong-1), SolidBody 1(M6 Tapped Hole1)(blenderaugershortonly-1), SolidBody 1(M6 Tapped Hole1)(blenderseat-1), SolidBody 1(Chamfer2)(blocker-1), SolidBody 1(Extrude2)(blocker2-1), SolidBody 1(M4 Tapped Hole1)(outbearingseat-1), SolidBody 1(Extrude6)(shaft-1), SolidBody 1(Extrude3)(sleeve7-1)
	Curve Data:N/A	



Loads and Fixtures

Fixture name	Fixture Image	Fixture Details
Fixed-1		Entities: 2 face(s) Type: Fixed Geometry

Resultant Forces				
Components	X	Y	Z	Resultant
Reaction force(N)	53.3007	44.7518	54.4406	88.3599
Reaction Moment(N.m)	0	0	0	0

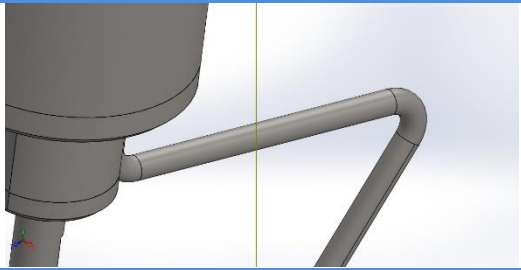
Load name	Load Image	Load Details		
Pressure-1		Entities:	1 face(s)	
		Type:	Normal to selected face	
		Value:	10,000	
		Units:	N/m^2	
		Phase Angle:	0	
		Units:	deg	
Torque-1		Entities:	2 face(s)	
		Reference:	Face< 1 >	
		Type:	Apply torque	
		Value:	2.2 N.m	

Connector Definitions

No Data



Interaction Information

Interaction	Interaction Image	Interaction Properties
Global Interaction		Type: Bonded Components: 1 component(s) Options: Independent mesh

Mesh information

Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Maximum element size	0.383237 in
Minimum element size	0.0191618 in
Mesh Quality	High
Remesh failed parts independently	Off
Reuse mesh for identical parts in an assembly (Blended curvature-based mesher only)	Off

Mesh information - Details

Total Nodes	486188
Total Elements	295910
Maximum Aspect Ratio	21.753
% of elements with Aspect Ratio < 3	97.2
Percentage of elements with Aspect Ratio > 10	0.11
Percentage of distorted elements	0
Time to complete mesh(hh:mm:ss):	00:00:56
Computer name:	HASSAN

Sensor Details

No Data



Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	91.5508	46.3413	98.5253	142.254

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Free body forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	0	0	0	0

Free body moments

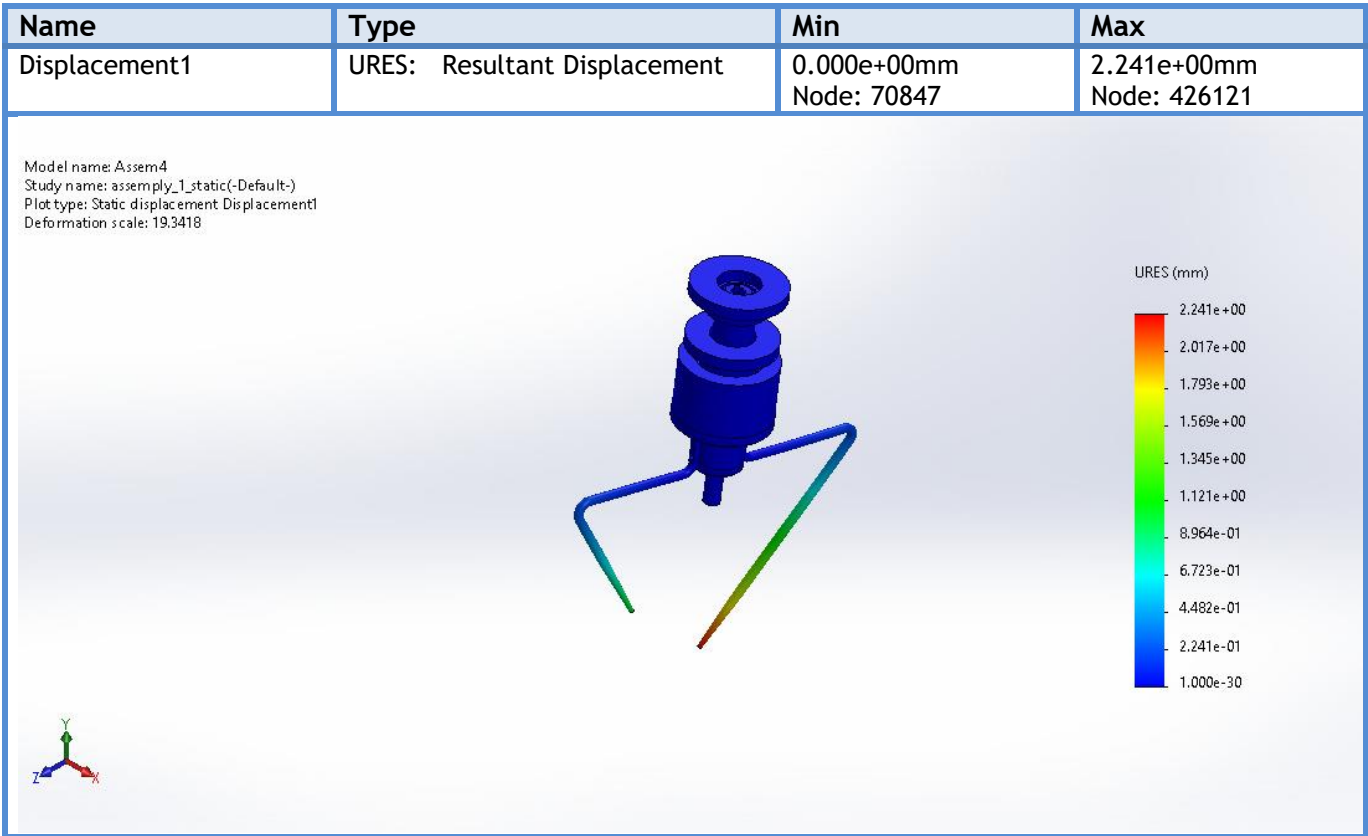
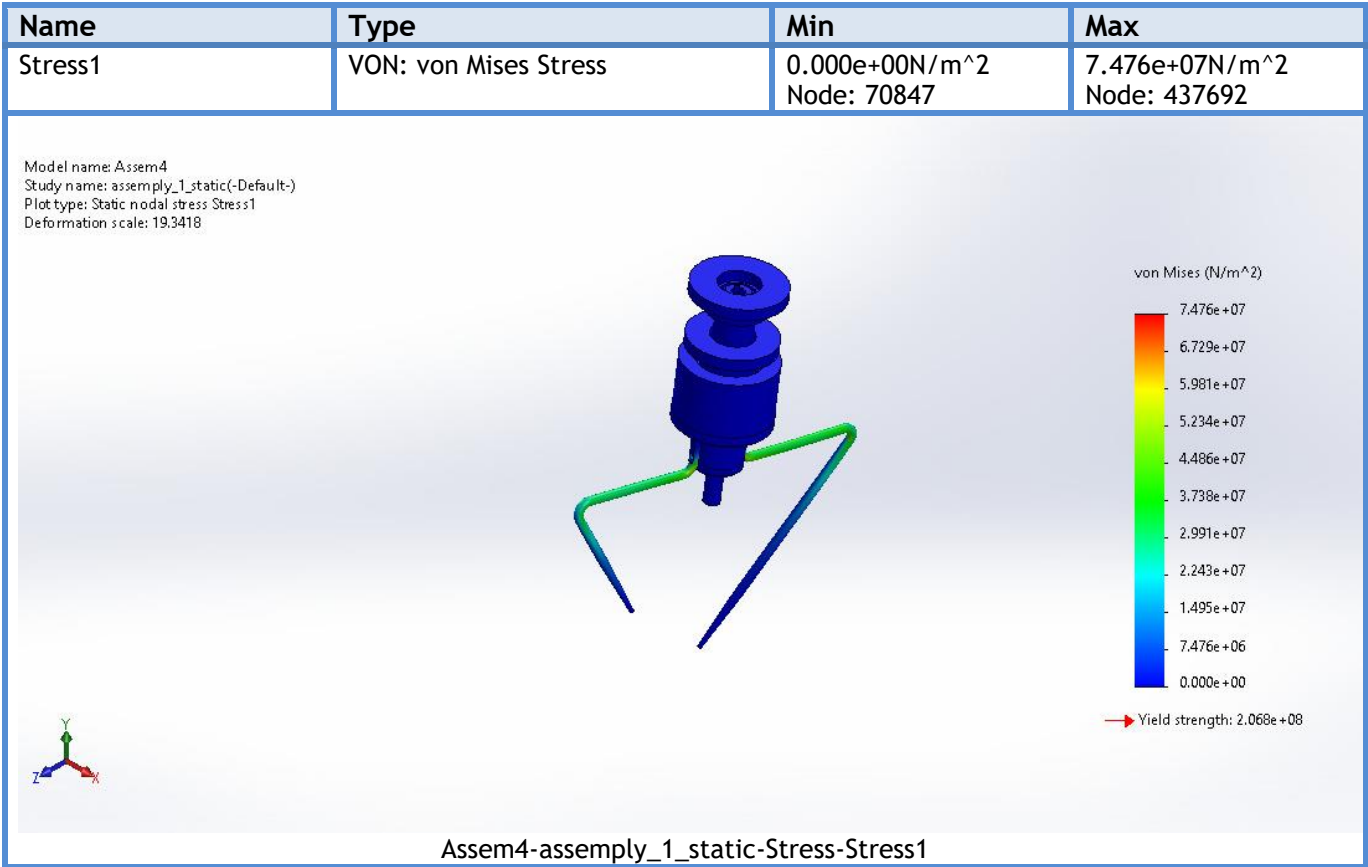
Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Beams

No Data



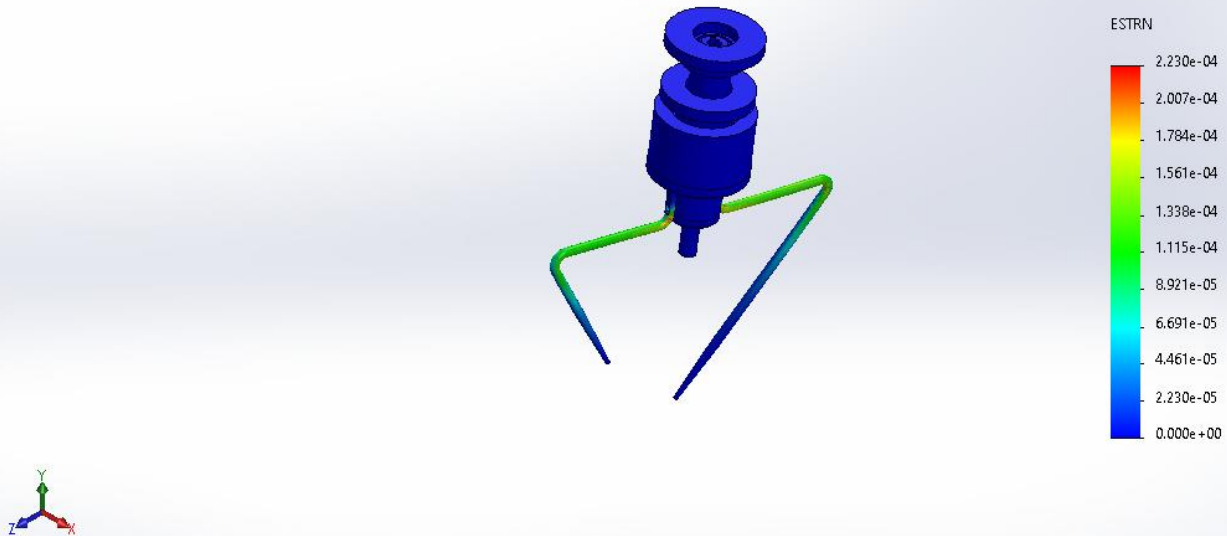
Study Results



Assem4-assemly_1_static-Displacement-Displacement1

Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	0.000e+00 Element: 42691	2.230e-04 Element: 267660

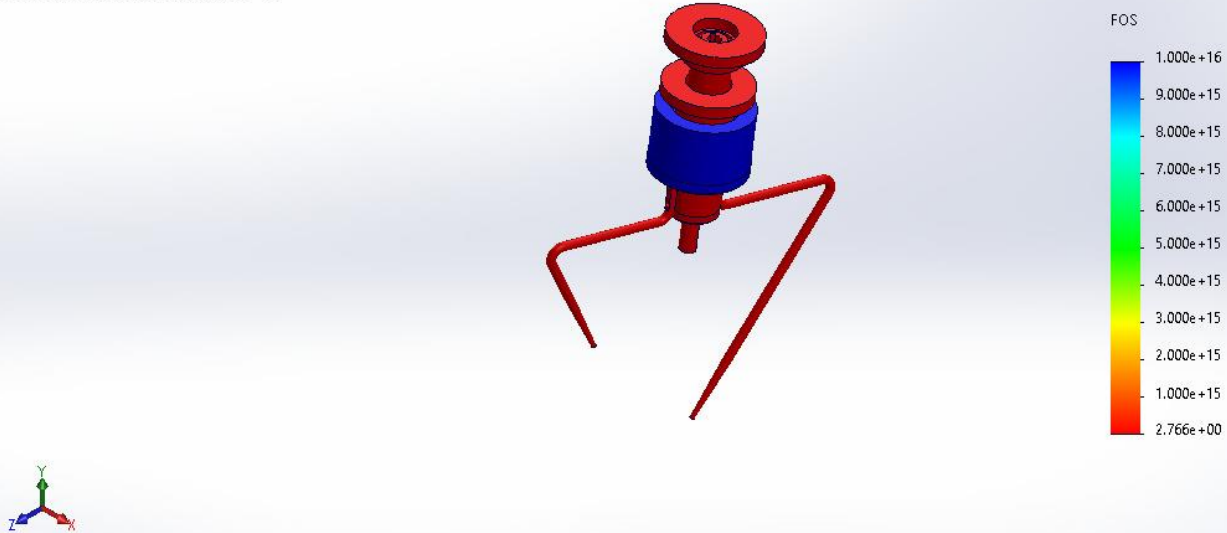
Model name: Assem4
Study name: assembly_1_static(-Default-)
Plot type: Static strain Strain1
Deformation scale: 19.3418



Assem4-assemly_1_static-Strain-Strain1

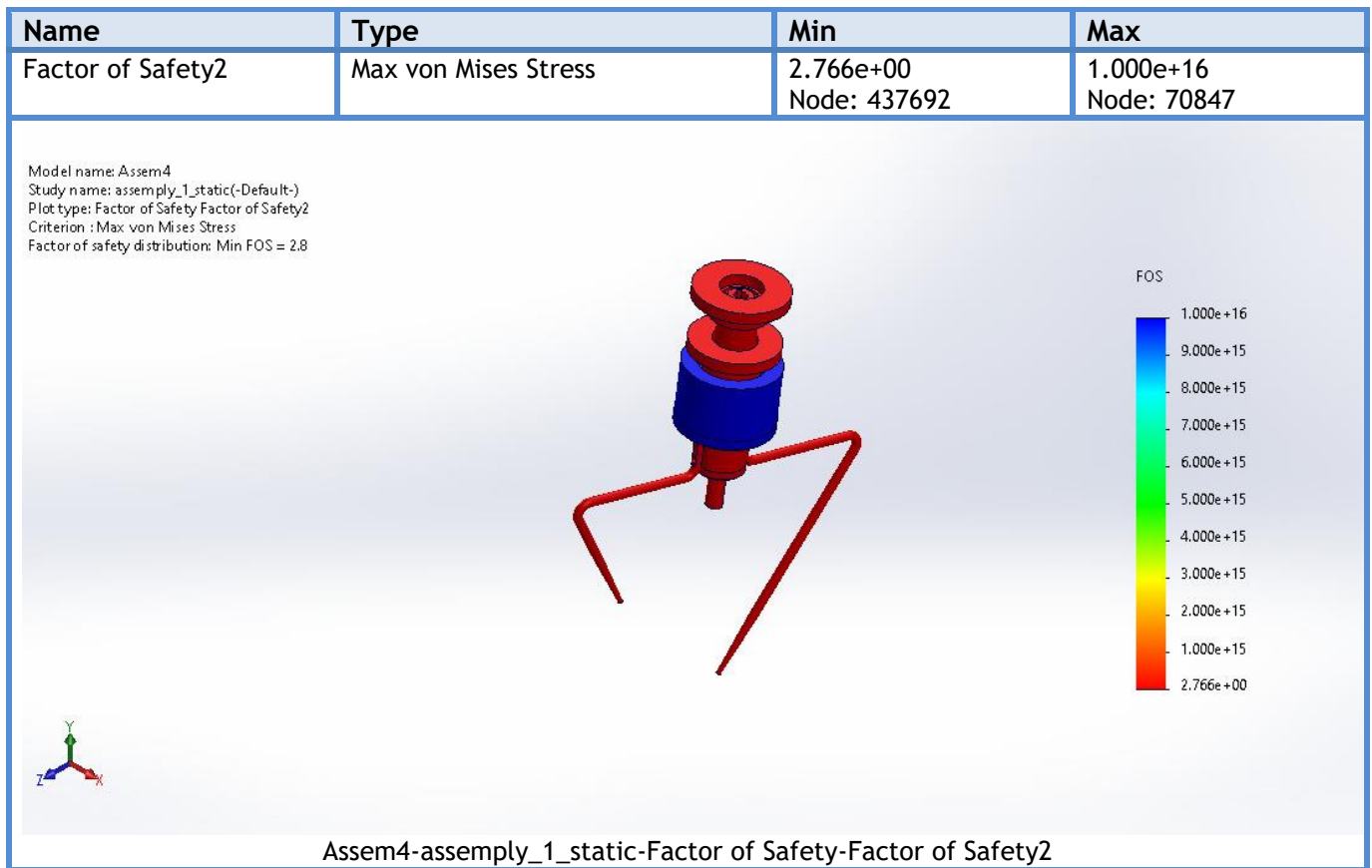
Name	Type	Min	Max
Factor of Safety1	Max von Mises Stress	2.766e+00 Node: 437692	1.000e+16 Node: 70847

Model name: Assem4
Study name: assembly_1_static(-Default-)
Plot type: Factor of Safety Factor of Safety1
Criterion : Max von Mises Stress
Factor of safety distribution: Min FOS = 2.8



Assem4-assemly_1_static-Factor of Safety-Factor of Safety1





Conclusion

